Difference between DoS and DDoS Attacks

Quick summary

Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks both aim to make a target service unavailable. The primary difference is that a DoS comes from a single source, while a DDoS comes from many distributed sources (often a botnet).

Definitions

DoS (Denial of Service): Attack where a single machine or connection floods a target with traffic or exploits a vulnerability to make it unavailable.

DDoS (Distributed Denial of Service): Attack where multiple compromised machines (often thousands) are used together to overwhelm a target, making mitigation and attribution harder.

Side-by-side comparison

Aspect	DoS	DDoS
Source	Single attacker or single compromised machine	Multiple distributed attackers/botnet machines
Scale & Volume	Usually lower volume; limited by single machine bandwidth	Very high volume; amplified by many machines
Complexity	Simpler to execute	More complex — requires control of many mac
Detection difficulty	Relatively easier to trace and block (single IP)	Harder to detect and trace due to many IPs and
Mitigation difficulty	Easier (block IP, rate-limit)	Harder — requires scrubbing, large-scale filteri
Common vectors	SYN flood, UDP flood, HTTP flood, resource exhaustion	Same vectors at larger scale; amplification atta
Attribution	Easier to attribute	Difficult due to spoofing and intermediaries
Example impact	Temporary downtime for small services	Major outages, global-scale disruption for large

Typical detection indicators

- Sudden spike in traffic or connections - Increased error rates (502/503) - Unusual traffic patterns from many IPs or geo-locations - Amplified request sizes or repeated identical requests

Common mitigation strategies

1. Rate limiting and filtering at the edge 2. Use of CDN and DDoS protection providers (scrubbing centers) 3. Upstream ISP collaboration and traffic blackholing when necessary 4. Network and application hardening (patching, resource quotas) 5. Use of WAFs, load balancers, and anomaly detection systems 6. Implement redundancy and failover to reduce single points of failure

Quick incident-response checklist

- Identify symptoms (traffic spike, logs) - Collect logs and packet captures for analysis - Contact upstream providers/CDN - Apply filtering rules and ACLs temporarily - Engage DDoS mitigation service if available - Post-incident: analyze root cause and update defenses

Historical examples (short)

Notable DDoS incidents include attacks against major DNS providers and large websites which caused widespread outages. DoS attacks historically targeted smaller services or individual servers.

Legal & ethical note

Launching DoS/DDoS attacks is illegal in most jurisdictions and unethical. The information in this document is for defensive, educational, and incident-response purposes only.

Further reading

- Network security textbooks and RFCs on TCP/IP - Vendor documentation for DDoS protection services - CERT advisories and blog posts from security teams

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